

NHANES Mini Project — Cleaned & Organized Version

1. Introduction

This project analyzes a small subset of NHANES-like demographic and anthropometric data.

We perform:

- Data cleaning

- Feature engineering (BMI, Age Groups)

- Statistical summaries

- Visualizations

- Outlier detection (Z-score)

2. Loading the Data

```
import pandas as pd
df = pd.read_csv('nhanes_clean.csv')
```

3. Creating Age Groups

```
bins = [0, 17, 40, 60, 80, 150]
labels = ["0-17 (child)", "18-40 (young)", "41-60 (adult)", "61-80 (senior)", "81+ (elder)"]
df["AgeGroup"] = pd.cut(df["Age"], bins=bins, labels=labels)
```

4. Calculating BMI

```
df['BMI'] = df['Weight'] / (df['Height']/100)**2
```

5. Summary Statistics

Mean height & weight by age group

```
df.groupby("AgeGroup")[["Height", "Weight"]].mean()
```

Min & max values

```
df[["Height", "Weight"]].agg(["min", "max"])
```

6. Correlation Matrix

```
import seaborn as sns
import matplotlib.pyplot as plt

corr = df[["Age", "Height", "Weight", "BMI"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```

7. Visualizations

Age distribution

```
plt.hist(df["Age"], bins=10, color="skyblue", edgecolor="black")
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

BMI boxplots

```
sns.boxplot(x="AgeGroup", y="BMI", data=df, color="lightgreen")
plt.title("BMI by Age Group")
plt.show()
```

8. Outlier Removal (Z-Score)

```
df["z_BMI"] = (df["BMI"] - df["BMI"].mean()) / df["BMI"].std()
df_no_outliers = df[df["z_BMI"].abs() < 3]
```

Plot without outliers

```
sns.boxplot(x="AgeGroup", y="BMI", hue="AgeGroup",
            data=df_no_outliers, palette="pastel", dodge=False)
plt.title("BMI by Age Group (No Outliers)")
plt.show()
```

9. Conclusion

This project demonstrates:

- Cleaning and structuring health data
- Creating derived variables
- Statistical group comparisons

- Detecting outliers
- Building clear visualizations

This document is formatted in a clean, academic style suitable for presentation or portfolio use.